



Article

Classification of Tomato Harvest Timing Using an AI Camera and Analysis Based on Experimental Results

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Abstract: Smart agriculture has the potential to solve labor shortages and improve production efficiency and prices at the time of shipment. Predicting tomato yields during the cultivation period is crucial for planning shipment volumes and costs in advance. We propose technology that utilizes an AI camera to enable producers to predict yields more accurately, and we verify the effectiveness of the developed system through experimental validation. Specifically, an AI-recognition camera was developed, utilizing You Only Look Once (YOLO) to detect individual tomatoes. The detected tomatoes are analyzed for size using point cloud data. Moreover, the AI-recognition camera performs to classify ripeness based on hue. This technology can achieve accurate ripeness classification without being dependent on the brightness of the greenhouse. To evaluate this AI classification camera, the predicted yield obtained from the camera was compared with the actual harvested yield in the field. The analysis showed an error rate of 6.85%, demonstrating sufficient accuracy for practical implementation. By introducing this system, efficient yield prediction can be achieved, leading to reduced labor costs, stable tomato supply, improved quality, and optimized market distribution. As a result, it is expected to contribute to the benefits of both shippers and consumers.

Academic Editor: Andrea Pezzuolo

Received: 16 January 2025

Revised: 14 February 2025

Accepted: 14 February 2025

Published: 19 February 2025

Citation: Okabe, Y.; Hiraguri, T.; Endo, K.; Kimura, T.; Hayashi, D. Classification of Tomato Harvest Timing Using an AI Camera and Analysis Based on Experimental Results. *AgriEngineering* **2025**, *7*, 48. <https://doi.org/10.3390/agriengineering7020048>

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Keywords: smart agriculture; LiDAR; point cloud; HSV color space; harvest prediction

1. Introduction

Issues that need to be solved in the social context of agriculture include labor savings and increased production volume owing to a decrease in the number of agricultural workers. Smart agriculture is a means to solve this problem, and agricultural methods that use Artificial Intelligence (AI), robots, and Internet of Things (IoT) sensors are attracting attention.

This study proposes a technology to support fruit harvesting using greenhouse-grown tomatoes as the target crop. By estimating the yield in advance, tomato growers can easily make appropriate supply plans for the markets and retailers to which they ship tomatoes. It also reduces the risk of over or under-supply relative to the demand, thus ensuring a stable supply. For example, if a large harvest is expected, proactive measures can be taken, such as arranging for processing and storage, to avoid a drop in market prices. Additionally, the labor and machinery schedules required for harvesting can be adjusted in advance to minimize waste costs and time. This proactive planning can also

help reduce waste and deterioration losses by, for example, using surplus tomatoes for processing. Thus, advanced yield estimation leads to cost reduction and profit maximization for growers and benefits shippers and consumers through a stable supply and improved quality.

Various approaches have been used in the technology and research related to tomato harvest predictions. For example, spectral images captured by a spectral camera have been analyzed using machine learning [1], and images have been analyzed using a machine learning Mask-CNN (Convolution Neural Network) to accurately determine the fruit-bunch condition and identify the differences in RGB [2,3]. Although these related studies have used methods to determine the degree of ripeness based on the color of tomatoes by identifying the RGB values of the captured images, it remains challenging to accurately identify images due to varying light conditions inside a facility. On the other hand, a method has been proposed to correct the discrepancy based on the average number of days of harvest by quantitatively analyzing data such as total temperature, sunshine hours, and CO₂ concentration, which growers have conventionally used to predict the harvest [4]. In related research, it is necessary to obtain data on many conditions, such as when sudden temperature changes occur or when it is difficult to obtain data on the growing environment, which is a significant hurdle in practical applications.

Tomatoes are a high-demand and popular crop. They rank first in global crop production, with 1.8 million tons of tomatoes produced annually, followed by onions (1.04 million tons) and cucumbers (0.91 million tons) [5]. Because tomatoes are the most popular crop, they were selected as the target for this study.

In recent years, AI and machine learning-based harvest prediction technologies have attracted attention, especially in tomato cultivation. This technology analyzes environmental data and crop growth conditions within greenhouses to predict yield and harvest timing. This allows farmers to easily plan their supply, preventing oversupply and excess inventory. AI technology is expanding beyond tomatoes to other crops such as strawberries, cucumbers, and lettuce. For these crops, prediction using AI-based image analysis and environmental data is advancing [6,7]. For example, in strawberries and cucumbers, harvest prediction using AI has already been implemented. In particular, for strawberries, AI analyzes color, size, and ripeness to predict the optimal harvest time. In addition, prediction technologies based on temperature, humidity, and soil components have been developed for cucumbers and lettuce [8,9]. This allows farmers to optimize harvest operations and maintain quality while ensuring a stable supply.

In this study, we propose a technology for predicting the shipping period by acquiring information on the size and ripeness of tomatoes. Furthermore, this technology aims to solve problems based on the direct requests of farmers, who were the inspiration for this research, as well as the background of the world, and it is highly necessary not only from an academic perspective but also from the perspective of social implementation, and this is the main motivation for this study.

2. Proposed Technology and Operational Procedure

This study established the following conditions for tomato harvest prediction with further operational considerations:

- A. Predicting tomato yield up to one week later;
- B. Identifying the right size for harvesting;
- C. Environment-dependent, light intensity-independent color discrimination of ripeness.

Based on these conditions, we developed an AI-discriminating camera that uses the machine learning You Only Look Once (YOLO) [3,4] to detect tomato objects and extract the size of the detected tomatoes from the image information. The size of the detected

tomatoes was extracted from the image information. Additionally, from the image data, a color-based method was used to identify the degree of ripeness independent of light intensity. Figure 1 shows the overall system configuration using these mechanisms. The system was operated as shown in Figure 2, and a demonstration experiment was conducted using the actual work of the harvester and the manual operation procedure of the developed system. The experimental site was located in Ama-gun, Aichi Prefecture, Japan, and the experimental tomato variety was “Karen” and was grown in a greenhouse. Figure 3 shows an overall map of Japan and an enlarged map of Aichi Prefecture where the experiment was conducted. The area indicated by the arrow is Ama-gun. The grower harvests tomatoes in the field by loading the containers onto a cart. The cart was equipped with an AI identification camera that collected image data when the tomatoes were harvested. These image data were used to predict harvest several weeks later. The image data were transferred to an external network cloud via Wi-Fi. The cloud data were analyzed to predict the yield.

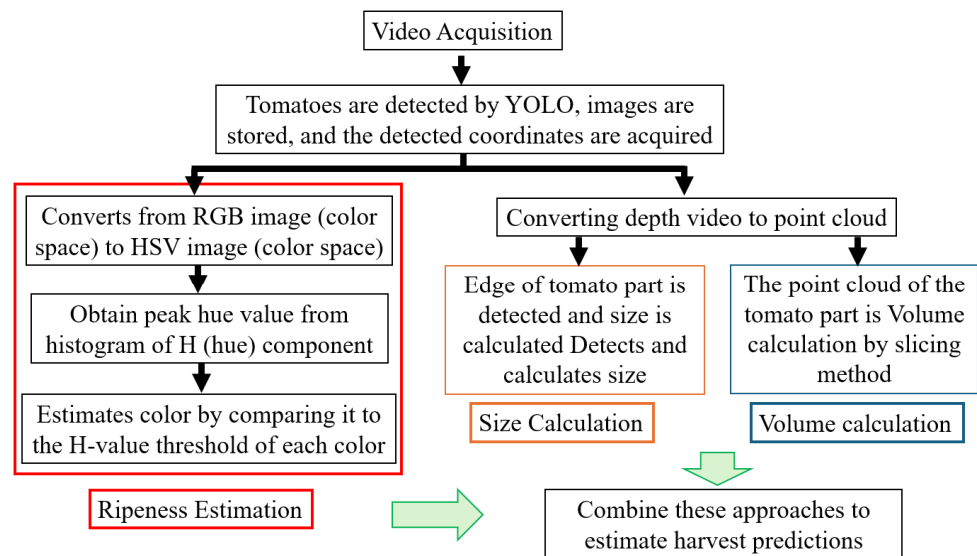


Figure 1. Flowchart of the harvest forecasting system.

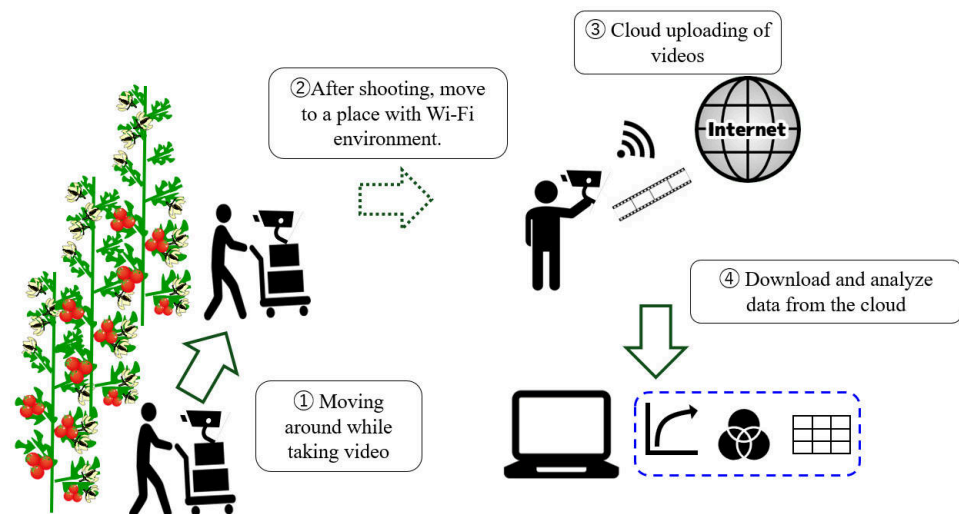


Figure 2. Procedures for system operation.

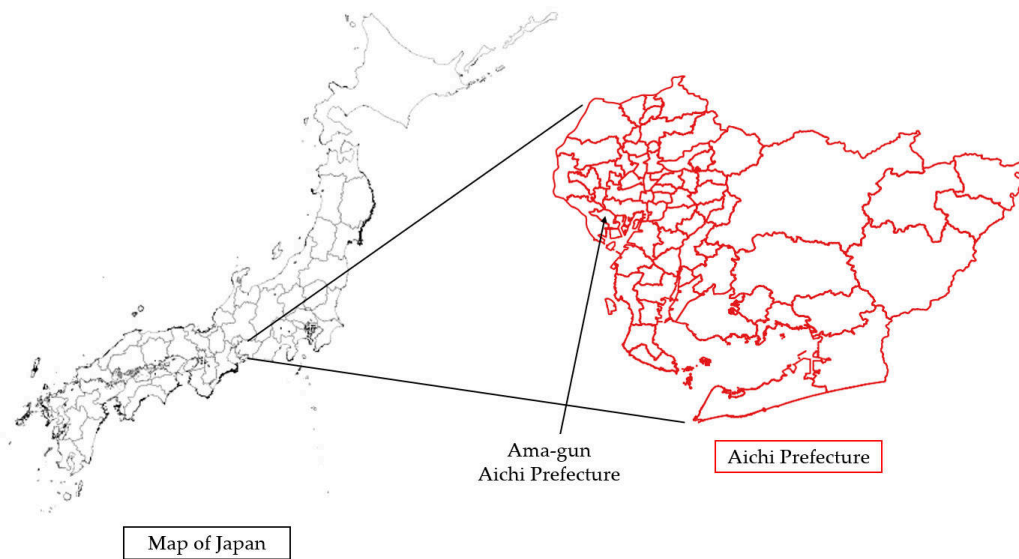


Figure 3. Experiment location.

3. Related Works and Issues

Based on the flowchart in Figure 1, we discuss the investigation and issues of each function of the tomato object verification, size measurement, and ripeness color classification technologies.

3.1. Challenges in Object Detection Technology

To predict the harvest, it is first necessary to detect tomatoes in camera images. Simple methods of detecting objects in images include edge detection, template matching, and machine learning. Template matching, including tomato size measurements, is discussed in Section 3.2. Machine learning includes Region-based Convolutional Neural Networks (R-CNN), Single Shot MultiBox Detector (SSD) [10], and YOLO. In this study, we used YOLO, which is the most popular and has a relatively fast detection speed and high detection accuracy [11].

Figure 4 illustrates the detection process using YOLO. The input image is divided into an $N \times N$ grid, as shown in Figure 4A. Subsequently, as shown in (B), each grid cell predicts the presence of an object in that cell. This includes the coordinates of the bounding box and the type of object. A confidence score was calculated for the bounding box predicted for each grid cell. This score represents the probability of the object being inside the box and its accuracy. Because multiple bounding boxes are predicted for the same object, the most reliable box is specified, and duplicate boxes are removed. The location of the detected object and its class label are finally outputted, as shown in Figure 5 [12–14]. YOLO has a low detection accuracy because false positives increase with low image quality or noise, as shown by the blue box in Figure 5 [11,15]. To improve the accuracy of tomato detection using YOLO, preprocessing such as resizing and random cropping based on the depth at the time of image capture is required, as well as optimizing pixels disturbed by moving speed and other factors. Annotation is also required to optimize the pixels disturbed by moving speed and other factors. Thus, it is necessary to solve the detection accuracy problem for operational applications.

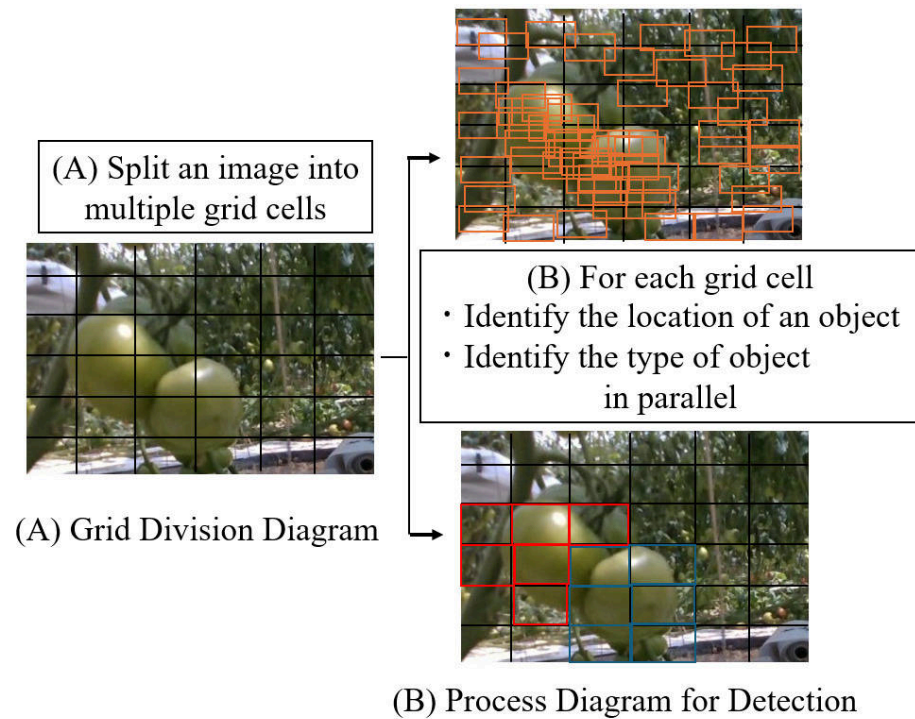


Figure 4. YOLO detection process.

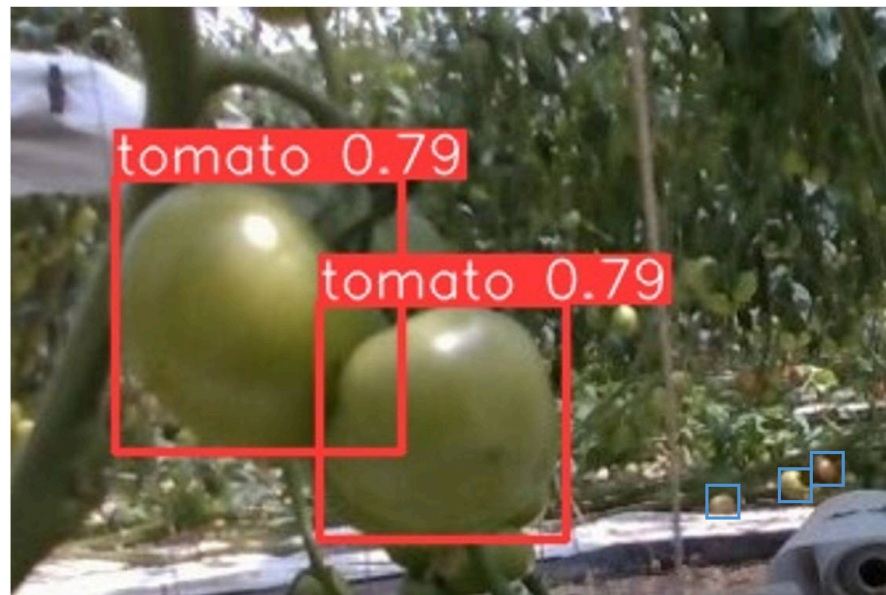


Figure 5. Difficult to detect by YOLO.

3.2. Tomato Size Measurement

Template matching is a simple method for size measurement. Related research in template matching includes implementing a multi-scale template matching method on Graphics Processing Units (GPUs) to reduce computation time [16], utilizing GoogLeNet to improve computation speed [17], and devising a new similarity measure between templates and images, Global-Aware Diversity (GAD), to improve matching accuracy [18]. GAD is a new similarity measure between templates and images that improves matching accuracy [19]. These are studies on improving the performance and application of

template matching technology, and methods and knowledge applicable to crop size measurements are also provided. As a concrete example, as shown in Figure 6, there is a method to extract the edges of an image obtained from binarization and template matching and measure the size based on the length and width of the edges. The shape of a tomato was recognized from the binarized image by template matching. The length and width of the tomato fruit and the thickness of the stem were calculated by counting the number of pixels in the white image. In other words, the size of a tomato can be determined by counting the number of pixels and comparing them using a ruler. However, setting up a yardstick within the angle of view was necessary to measure the actual size. The actual size of an object was known. Thus, the size of a pixel was calculated from the number of pixels in the object. The size was then multiplied by the number of pixels in the tomato area, as determined by binarization, to calculate the size. This requires the use of a yardstick. Table 1 compares the actual measured sizes and the sizes analyzed using template matching. Compared to the actual measurements, the measurement results were highly accurate, with an error rate of less than 10%. However, this method is impractical because it requires constantly installing a yardstick for operational purposes.

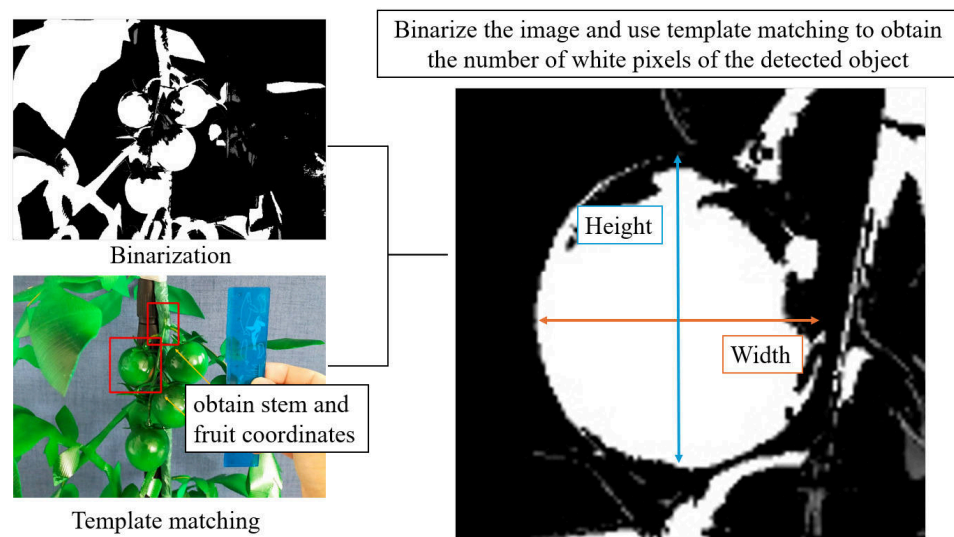


Figure 6. Image processing.

Table 1. Accuracy by template matching.

Measurement Method	Height	Width
Measured value [mm]	30.0	35.0
Calculated value [mm]	27.6	38.0
Error ratio [%]	8.00	8.57

Other measurement methods include Light Detection And Ranging (LiDAR) and Depth Cameras [20,21]. However, when the resolution is low, these methods struggle to achieve accurate measurements. Additionally, the high-analysis processing load poses a challenge. While there have been research results for size measurement in static environments, the dynamic environment in which data are acquired while moving during this research operation proves to be unstable, making practical applications difficult [22,23]. Therefore, we propose a simpler method that can be implemented in dynamic environments.

3.3. Determination of Ripeness and Harvest Prediction

The ripeness of tomatoes must be estimated to predict their harvest. The ripeness of tomatoes can be distinguished by a color change in the epidermis of the fruit: blue (un-ripe), green (ripening), and red (fully ripe). The color can be identified using the RGB components of the tomato's surface or by converting the color change into an index, such as the Hue Saturation Value (HSV). Another method is spectral analysis, which measures the reflectance of a surface at specific wavelengths using a spectrometer. Additionally, methods have been proposed to integrate and discriminate between RGB and spectral data using machine learning.

An RGB camera is useful because it is relatively inexpensive, readily available, and has a low processing load [24–26]. However, they are easily affected by ambient light (brightness and color temperature), and surface gloss and shadows interfere with accurate color determination. Therefore, color discrimination using sunlight, such as in greenhouses, is extremely difficult to achieve. Spectral analysis [27,28] is less affected by ambient light because it uses specific wavelengths. However, spectrometers and spectral analysis equipment are expensive, require calibration for initial installation, are complicated to operate, and have a large processing load. Machine learning [29] is beneficial because it is versatile; however, it is difficult to put it into practical use because it requires a large amount of data collection for learning and relearning when the light intensity or camera settings change. On the other hand, HSV [30,31] is capable of making judgments that are close to those made by human senses and can quantify color differences by using hue, which is less affected by changes in light intensity than RGB and is sufficient if ripeness is judged only by color. Based on this background, this study proposes using HSV for color discrimination of ripeness.

4. Detection of Tomatoes in the Field

4.1. Method of YOLO Detection

As mentioned previously, YOLO was used to detect tomatoes. To set up the system, we installed libraries such as ultralytics, opencv-python, and numpy in a Python 3.9 environment. After that, we created our own training data for a dedicated tomato detection model. The process of creating training data consists of preparing tomato images and labeling them with annotations. In this study, we used the labelImg tool for annotation. After all images have been labeled, training was performed. The dataset for training is divided into training and test datasets, and the configuration information is described in the data.yaml file. Next, we describe the preprocessing and training parameters. The image data used in YOLO were actual images captured in the field. First, multiple images of the object to be detected were prepared. In this study, 1166 images were obtained. To enhance detection accuracy even in the case of overlapping objects, low pixel counts, blurred objects, or overlapping leaves, as shown in Figure 5, we prepared images in various states. Next, as shown in Figure 7, the objects to be detected are surrounded by red arrows in the image and labeled with annotations. In this study, only tomatoes were annotated to detect tomatoes because the model was intended for tomato detection. The class label, in this case, was "tomato". After all images were annotated, the last step was training. The training parameters were set to 315 test images, 100 training cycles, and a batch size of 32. Because it is common for the number of test images to be 2–30% of the number of training images, we set the number of test images to 315, which is approximately 30% of the 1166 images. The number of training sessions was set as 100 to avoid overtraining. The batch size was set to 32 [32,33].

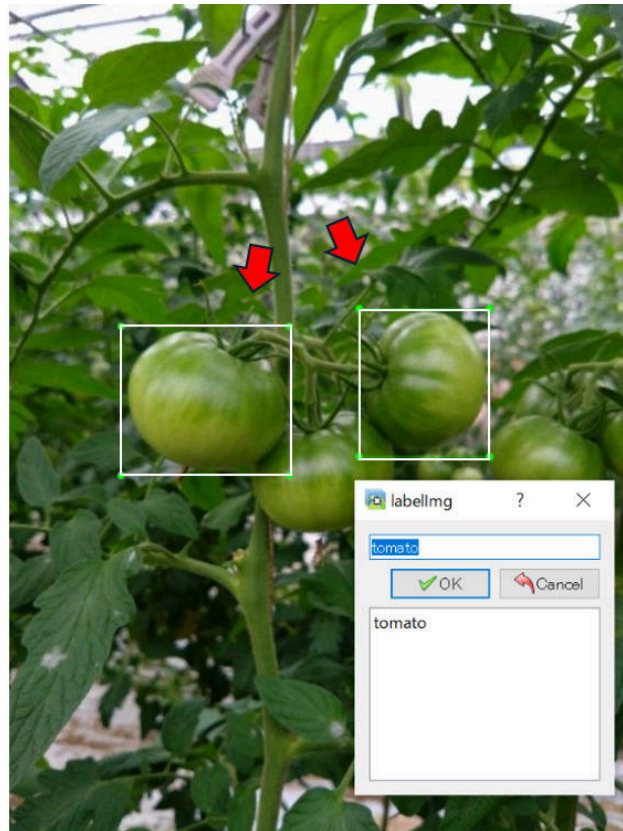


Figure 7. YOLO preprocessing method (annotation).

4.2. Results of Detection Method

The accuracy of the tomato detection method constructed using YOLO was confirmed. During the experiment, a camera was mounted on the cart, with the view angle directed toward the bottom of the cart. The time required for each row was approximately 1 and 30 s to 2 min. Figure 8 shows how YOLO detects tomatoes within the range of the view angle during the actual shooting. To avoid duplicate detections of the same tomatoes, tracking was performed to avoid quantity errors [34]. In other issues, the tomatoes circled in red in Figure 8 were excluded from harvesting in another row (the row in the back). The excluded tomatoes were also included in the quantification, which resulted in errors. Therefore, a threshold was set to exclude those with small bounding boxes, and the tomato boxes on the far side were adjusted so that they could not be detected [35].



Figure 8. Consideration of other-row tomato detection.

The evaluation method involves pausing the video at equal intervals and repeating the analysis within an image. Videos were recorded in the greenhouse during the experiment. The percentage of tomatoes among the objects detected in the video was defined as the percentage of correct responses. If the number of tomatoes detected correctly is a True Positive (TP), and the number of tomatoes detected incorrectly is a False Positive (FP), the correctness rate can be expressed as Equation (1). The detection rate is the ratio of the number of tomatoes in the video that the model can detect to the actual number of tomatoes in the video. Assuming that TP and the number of tomatoes present in the video are not detected as False Negatives (FN), the detection rate can be expressed as Equation (2) [36].

Table 2 shows these results. The correct response rate was 0.99, and the detection rate was 0.88. Figures 9 and 10 show the false positive and negative detections. The areas circled in red in Figure 9 are false detections, and the possible causes include the similarity in shape to tomatoes and the presence of unexpected objects that were originally projected. Figure 10 shows that the area circled in red is undetected, possibly because YOLO is not good at detecting small objects, and small objects are undetected. Other possible reasons include that the slight decrease in the detection rate is thought to be because the leaves were almost completely hidden, making it difficult to detect them.

Table 2. Correctness and detection rate results.

Accuracy rate [%]	99.00
Detection rate [%]	88.00



Figure 9. False detection.

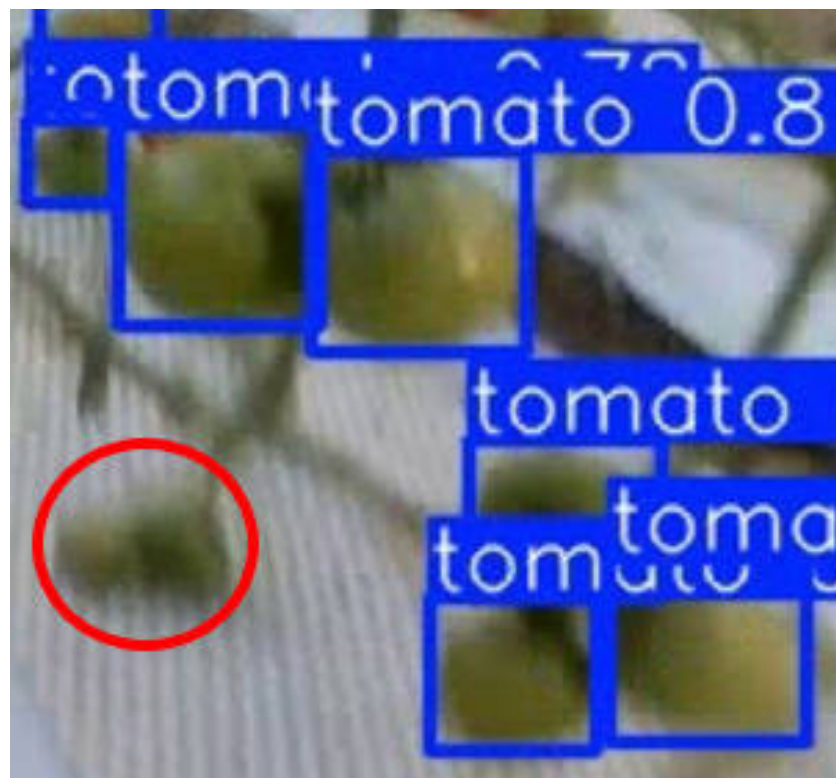


Figure 10. False negative detection.

As the correct response rate was 99%, 99% of the detected objects were tomatoes. The detection rate was 88%, indicating that 88% of the total number of tomatoes in the video were detected. Twelve tomatoes (12%) were undetectable. The 99% correct rate means that 99% of the objects detected were actually tomatoes, and the remaining 1% detected objects that were not tomatoes. The detection rate is 88%, which means that 88% of the total number of tomatoes in the video were detected. The remaining 12% of tomatoes are not detected, meaning that they are not detected. In this study, we did not estimate the number of tomatoes shipped. Therefore, even if there are omissions, this is not a problem because it is possible to predict how much can be harvested.

The mean Average Precision (mAP) was used to evaluate the performance of the object detection models [37]. mAP was calculated from the Average Precision (AP), which was obtained by integrating the curve of a graph plotting the percentage of correct responses on the vertical axis and the detection rate on the horizontal axis. If r is the value of Recall and $Precision(r)$ as the corresponding accuracy rate, the AP is expressed by Equation (3). The mAP is obtained by averaging the AP by the label class. If C is the number of label classes and AP_i is the value of AP for label class i , mAP is expressed by Equation (4). The Intersection over Union (IoU) measures the degree of agreement between the predicted region and the correct region and is expressed on a scale from 0 to 1. The closer the value is to 1, the more accurate the prediction. mAP50 is an index that sets the threshold value to 0.5 or higher. Figure 11 shows a graph of the mAP50 for the results of this study. Figure 11 shows that mAP50 increased as training progressed. This indicated that the accuracy of the detection model improved with repeated training. As a mAP of 80% or higher is generally recognized as having sufficient accuracy [11,38], the discrimination accuracy of this training model is high.

$$\text{Accuracy rate} = \frac{TP}{TP + FP} \quad (1)$$

$$\text{Detection rate} = \frac{TP}{TP + FN} \quad (2)$$

$$AP = \int_0^1 Precision(r) dr \quad (3)$$

$$mAP = \frac{1}{C} \sum_{i=1}^C AP_i \quad (4)$$

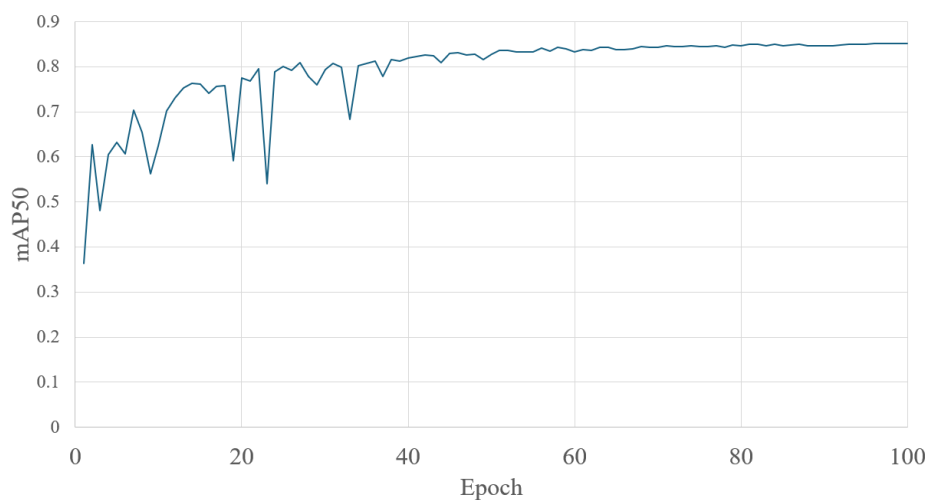


Figure 11. mAP50 change in epoch.

5. Tomato Size Measurement Method

5.1. Method of Distance and Size Measurement by LiDAR

The method for calculating size relative to distance is based on point cloud data. Point cloud data represents the three-dimensional configuration of an object using a collection of points with XYZ coordinates, as shown in Figure 12 [39,40]. In this research, a depth camera is used instead of a 2D camera, and the depth camera can acquire depth images. Each pixel in the depth image contains information indicating the distance to the object, and this information can be used to estimate the position of the object in 3D space. The coordinates corresponding to each pixel in the depth image are calculated, and the three-dimensional shape of the object is represented as a point cloud.

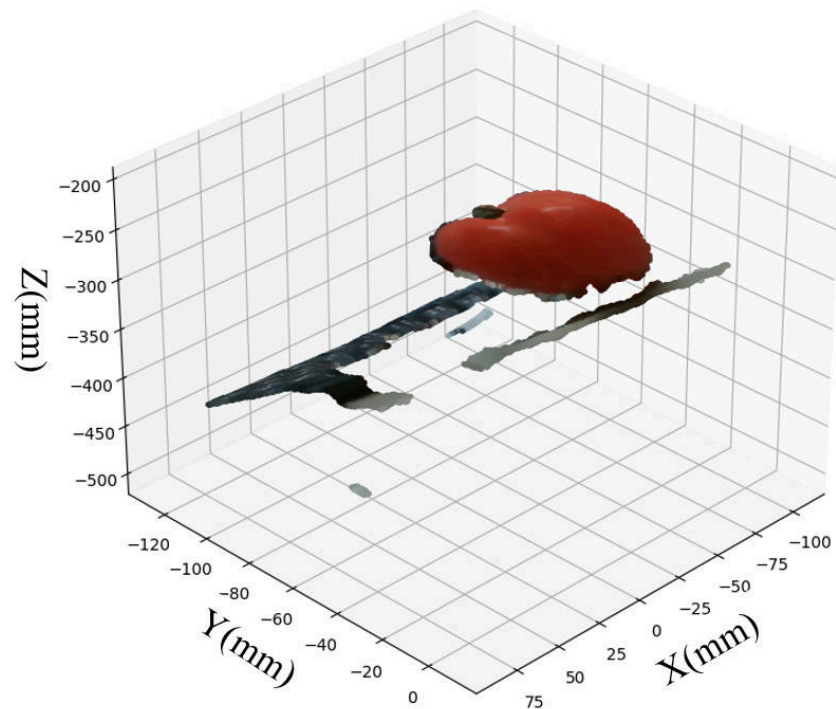


Figure 12. Image of point cloud.

To measure the size, the normal vector of a point cloud was used to obtain the point cloud of the tomato edge area. Figures 13 and 14 show that the normal vector indicates the direction perpendicular to the surface at a point on a certain plane or curved surface [41–43]. When measurements are taken on a horizontal surface, such as a desk, as shown in Figure 13A, the points appear on a plane, as shown in Figure 13B, and the normal vector is perpendicular to the plane. Thus, they all point in the same direction. Figure 14 shows the results for the curved surface. When measuring a curved surface such as a tomato, as shown in Figure 14A, a point cloud appeared on the surface, as shown in Figure 14B. The normal vector is perpendicular to each point cloud on the surface. Therefore, the direction varies from place to place.

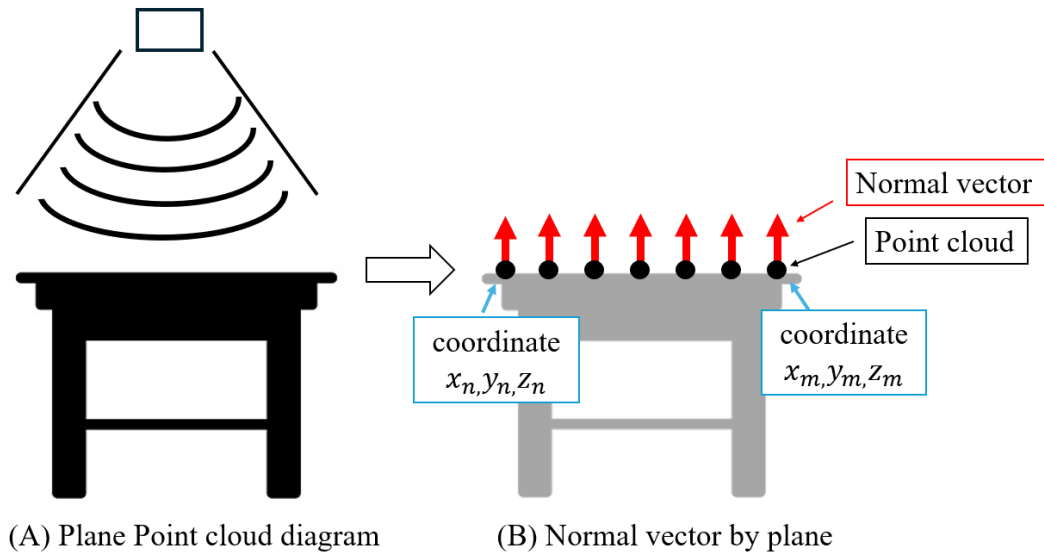


Figure 13. Normal vector image (plane).

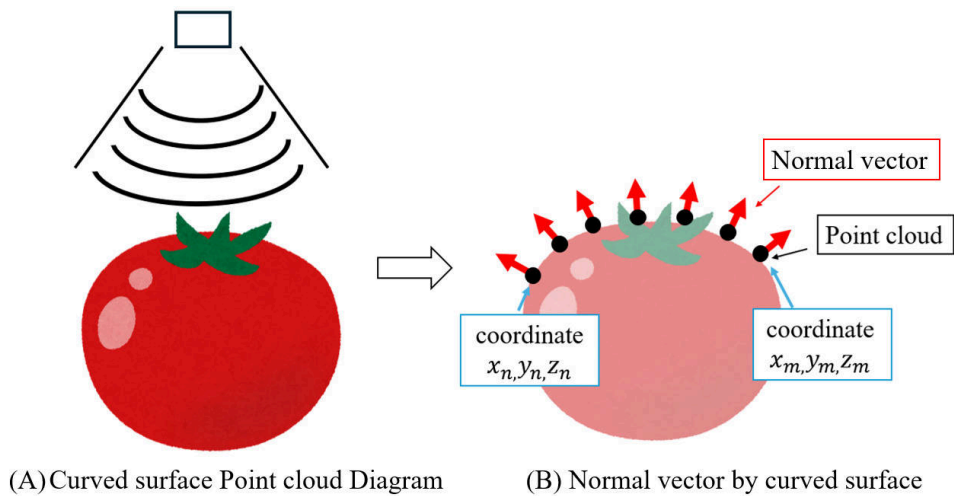


Figure 14. Normal vector image (curved surface).

Based on the characteristics of these normal vectors, the edge of the tomato surface area was detected, and the Euclidean distance D in 3-dimensional space was expressed as shown in Equation (5) using the point group coordinates $(x_n, y_n, z_n)(x_m, y_m, z_m)$ of the two points at the edge [44,45].

$$D = \sqrt{(x_n - x_m)^2 + (y_n - y_m)^2 + (z_n - z_m)^2} \quad (5)$$

5.2. Result of Size Measurement

In this experiment, we assessed size measurement accuracy by comparing it with template matching, using point cloud data to conduct multiple, repeatable, and quantitative evaluations. Size measurements were performed for each tomato in fields #1–#3, as shown in Figure 15. Table 3 shows the results of template matching measured by template matching 10 times, and Table 4 shows the results of measuring each tomato using a point cloud. It was confirmed that size measurement was possible without using a yardstick as in the template matching method.



Figure 15. In-field reproduction using point clouds and color images.

Table 3. Measurement accuracy by template matching.

	Fruit Height	Fruit Width
Actual tomato size [mm]	30.0	35.0
Experimental results [mm]	28.6	37.2
Error ratio [%]	8.00	8.57

Table 4. Tomato #1~#3: measurement results.

	#1	#2	#3
Height [mm]	35.5	21.0	20.4
Width [mm]	40.8	25.8	32.0

5.3. Method of Measurement of Volume by Point Cloud Data

To measure the volume, the half surface of a tomato was obtained as a point cloud, as shown in Figure 16, and the calculation was performed by integrating the cross-sectional area of the point cloud of the half surface [46]. The left image in Figure 16 shows the front view of the point cloud and color image, and the right image shows the side view of the point cloud. The volume of the entire tomato was calculated by dividing the tomato point cloud into multiple cross-sections by depth, as shown on the right, integrating the cross-sections, and doubling them, assuming the backside of the tomato. If the xy coordinates of the edge points are x_i and y_i , the cross-sectional area [mm²] $A(z)$ is expressed as shown in Equation (6). Using Equation (6) and assuming that the volume is V [mm³] and the number of cross-sectional area divisions is N , the calculation equation is expressed as Equation (7).

$$A(z) = \frac{1}{2} \left| \sum_{i=1}^n (x_i y_{i+1} - x_{i+1} y_i) \right| \quad (6)$$

$$V = 2 \int_0^N A(z) dz \quad (7)$$

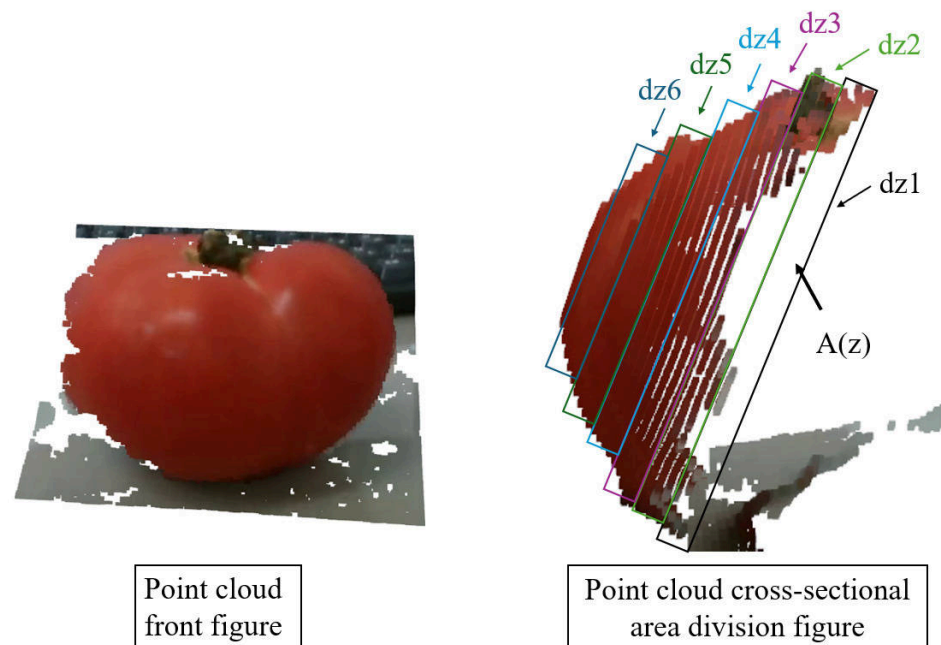


Figure 16. Volume calculation image.

5.4. Results of Measurement of Volume by Point Cloud Data

Three randomly sized tomatoes were prepared for accurate volumetric measurements, as shown in Figure 17. Table 5 lists the accuracies of each volumetric measurement. As shown in Table 5, tomatoes #1 and #2 have error rates of approximately 5% and 9%, respectively, whereas tomato #3 has an error rate of approximately 26%. This is because the shapes of tomatoes #1 and #2 are generally elliptical, so the edges can be detected. However, tomato #3 is slightly tilted and distorted compared with tomatoes #1 and #2, as shown in Figure 17.

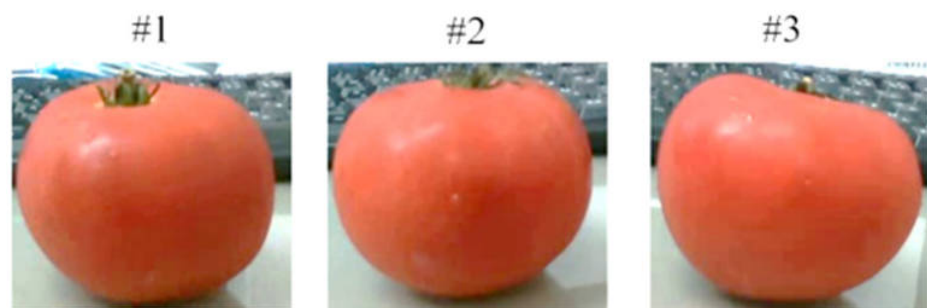


Figure 17. Comparison of different tomato shapes.

Table 5. Tomato #1~#3 volume measurement errors.

	#1	#2	#3
Actual volume value [cm ³]	94.7	107.1	76.2
Measured			
Volume value [cm ³]	90.0	97.2	96.0
Error ratio [%]	4.96	9.24	25.98

Therefore, edge detection using the point cloud was not successful. As shown in Figure 16, the volume was obtained by integrating the cross-sectional area. The accuracy of these measurements varied depending on the amount of segmentation. Table 6 lists the results of the volumetric measurements with 5, 10, and 20 slices. Table 6 shows that 20 slices yielded the highest accuracy. However, because the process becomes slow when the number of slices increases, the number of slices was set to 10 to strike a balance between accuracy and processing speed.

Table 6. Volumetric measurement results according to the number of divisions.

	#1	#2	#3
Number of Slices, 5 [cm ³]	103.9	125.7	101.8
Number of Slices, 10 [cm ³]	90.0	97.2	96.0
Number of Slices, 20 [cm ³]	91.0	100.8	90.8

6. Harvest Forecasts for Ripeness

6.1. Method to Classify Ripeness by Color

The HSV color space is a color model comprising hue, saturation, and value/brightness. When the light intensity, saturation, and lightness change, the hue is hardly affected by the light intensity, making it possible to obtain true colors [47–50]. As shown in Figure 18b, the hue represents a hue ring; therefore, the RGB values can be expressed by the angle of the hue ring.

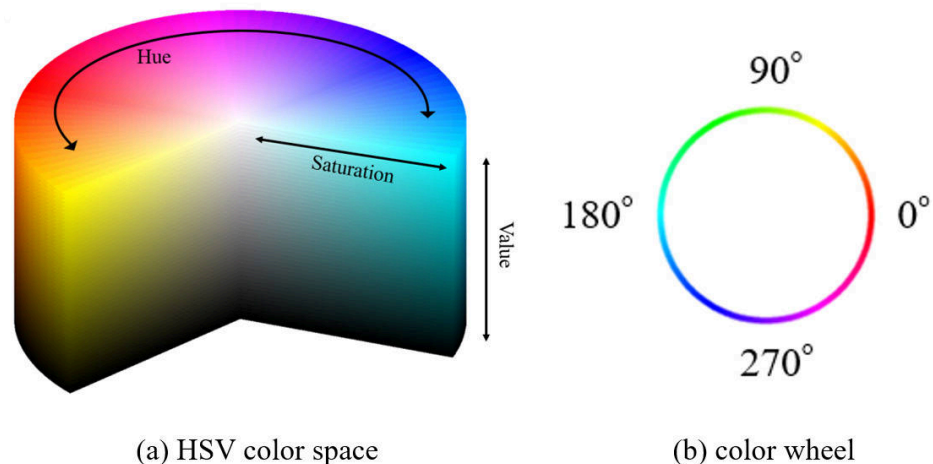


Figure 18. HSV color space overview.

The change in the RGB component values when the light intensity changes in the RGB color space is shown using Equations (8)–(12). The equations also explain why the conversion from the RGB color space to the HSV color space and the use of the RGB component values after the change in light intensity does not affect the hue in the HSV color space. First, when the light intensity changes, k is added to the component values (R , G , B) in the RGB color space, and the component values change proportionally, as shown in Equations (8) to (10). Equations (11) and (12) show the maximum and minimum values after a light intensity change (R' G' B').

$$R' = k \times R \quad (8)$$

$$G' = k \times G \quad (9)$$

$$B' = k \times B \quad (10)$$

$$C_{\max} = (R', B', G') \quad (11)$$

$$C_{\min} = (R', B', G') \quad (12)$$

To convert the RGB color space into the HSV color space, lightness was obtained using Equation (13), saturation using Equation (14), and hue using Equations (15)–(17). Note that formula (15) represents the maximum value of R' , formula (16) represents the maximum value of G' , and formula (17) represents the maximum value of B' . The brightness in Equation (13) changes in proportion to the amount of light intensity change, whereas the saturation and hue in Equations (14)–(17) are canceled out by the amount of light intensity change and are not affected by changes in light intensity [51–54].

$$V(\text{Value}) = C_{\max} \quad (13)$$

$$S(\text{Saturation}) = \frac{C_{\max} - C_{\min}}{C_{\max}} \quad (14)$$

$$H(\text{Hue}) = 60 \times \frac{G' - B'}{C_{\max} - C_{\min}} \quad (15)$$

$$H(\text{Hue}) = 60 \times \frac{B' - R'}{C_{\max} - C_{\min}} + 120 \quad (16)$$

$$H(\text{Hue}) = 60 \times \frac{R' - G'}{C_{\max} - C_{\min}} + 240 \quad (17)$$

6.2. Results of Hue Discrimination

The hue discrimination was evaluated using red, orange, and green tomatoes. To confirm the change in hue according to light intensity, we compared the results in dark and bright environments by changing the light intensity. The light intensity was set to 1200 cd in light and 5 cd in the dark. These were the maximum and minimum values of the variable light intensity range of the LEDs used in this experiment. Figures 19–21 show the cumulative hue values for each pixel within the red, orange, and green images. Identification was performed by selecting the peak value at which the hue was the highest.

For example, the red pixel in Figure 19 is three in the dark and zero in light. The orange color in Figure 20 is 11 under both dark and light conditions. Green in Figure 21 has a hue value of 30 in the light and 31 in the dark, and all colors have approximately the same hue value, regardless of the light or dark changes. Applying the hue values of each color to the color thresholds in Table 7 we observe that the hue values in Figure 19 are 3 and 0, which correspond to the 0–6 range for red, and thus are assumed to be red. Figure 20 shows that the hue value is 11, corresponding to 11–15 for orange. Figure 21 shows that the hue values are 30 and 31, corresponding to light green to yellow green, and thus, estimated to be in the light green to yellow-green stage.

Using this table as a reference, color discrimination was outputted as a judgment result based on the range of threshold values.

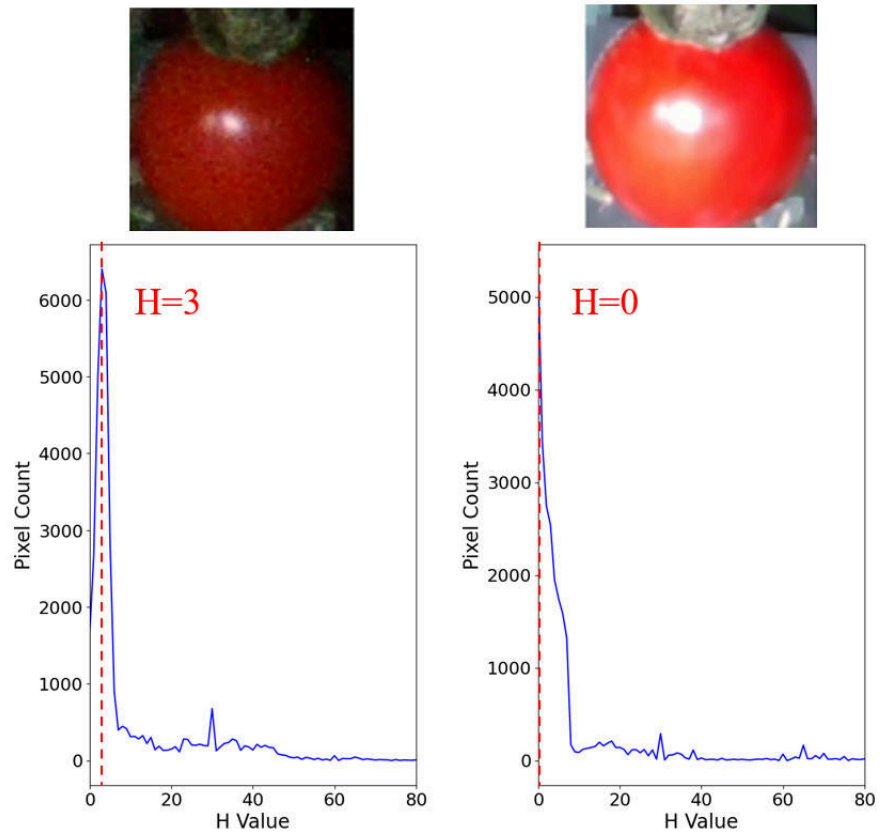


Figure 19. Hue results for red tomatoes.

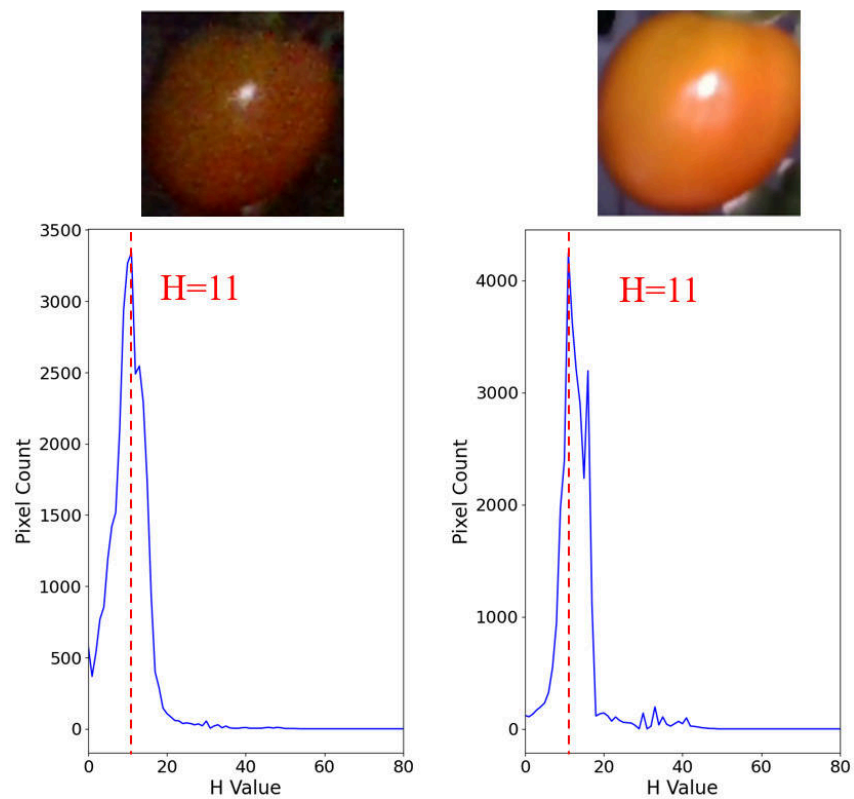


Figure 20. Hue results for orange tomatoes.

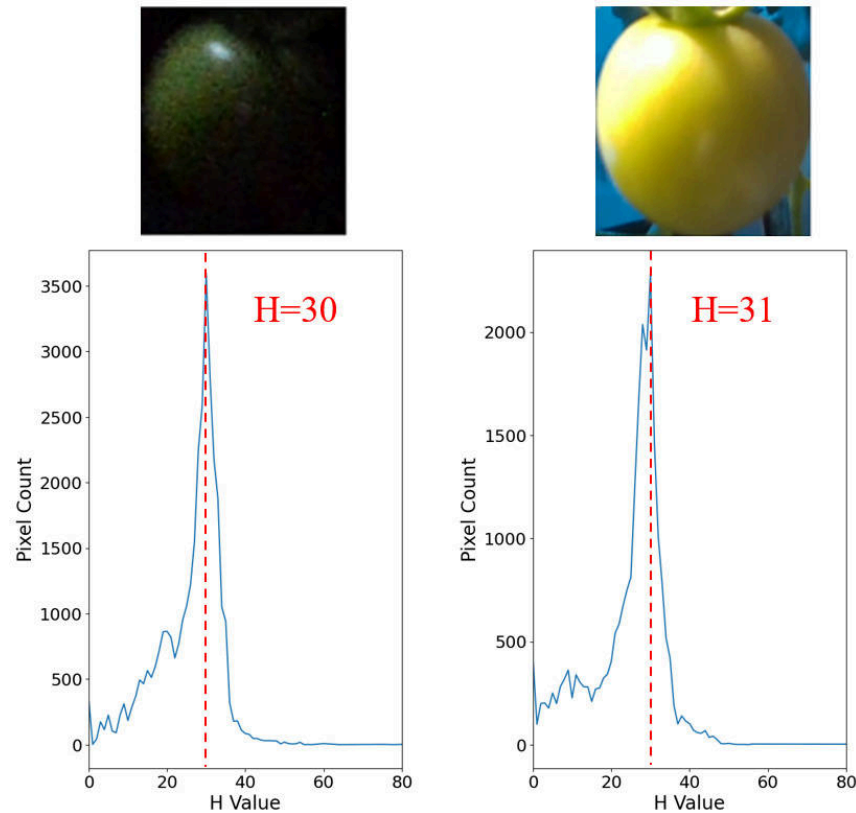


Figure 21. Hue results for green tomatoes.

Table 7. Threshold values for each color.

	Hue Value
Red	0~6
Orange to Red	7~10
Orange	11~15
Yellow Green to Orange	16~22
Yellow Green	23~27
Light Green to Yellow Green	28~32
Light Green	33~39
Green to Light Green	40~45
Green	46~60

7. Experimental Results and Harvest Predictions

A demonstration experiment is conducted by combining these technologies. Figure 22 shows the AI identification camera system used in the demonstration experiment. In this configuration, the LiDAR is installed in the upper part, and the angle of view can be adjusted using motor control. The lower part of the camera is powered by a battery, allowing for mobile operation without a power supply. The continuous photographing time was approximately 5 h. The display shows the remaining battery charge so that the operator can check it while working. The experimental system was mounted on a harvest cart, and future harvest forecasts were made simultaneously with harvesting. Figure 23 shows a scene from this experiment. The experiment aimed to predict the harvest yield for up to one week. The accuracy of the AI-discriminating camera system was evaluated by analyzing the captured data and comparing the analysis results with the actual harvested quantity. The analysis outputs the size and color of the tomatoes as the identification results.

During the grower's operation in the rented demonstration field, only the ripeness color visible to the human eye was harvested. Consequently, this experiment focused solely on the color of ripeness. Each row was approximately 90 m long, with approximately 100 plants per row. Imaging was performed using an AI-recognition system while moving at a constant speed of approximately 0.5 m/s. The image comprises four rows. Table 8 presents the actual harvested quantities for each row. It also shows the total values of the analysis results categorized as orange, orange to red, and red, along with the error rates for each row. The overall error rate for all four rows was maintained within 12%, with an average error rate of approximately 6.85%. This indicates that there is not much difference between the ripeness seen by the human eye and the ripeness analyzed, which confirms that the developed AI-recognition camera demonstrated sufficient performance for practical applications in the field.

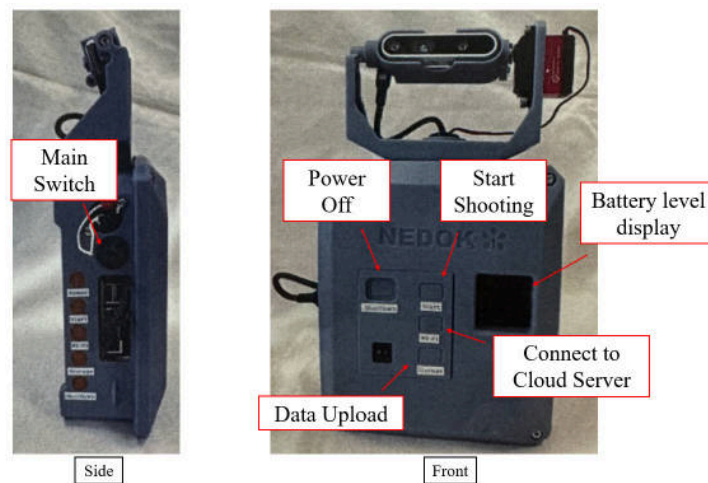


Figure 22. Development of an experimental device implementing an AI camera.



Figure 23. Experimental view.

Table 8. Error rates with actual yield analysis results.

	1 Row	2 Row	3 Row	4 Row
Actual harvest	106.0	84.0	81.0	105.0
Analysis results	106.0	78.0	75.0	93.0
Error rate [%]	0	7.14	7.41	11.40

Finally, the method for estimating harvest quantities one week prior to the harvest period was examined. Table 9 presents the actual harvest amounts recorded one week after the analysis. Here, “Results of analysis (A)” represents the combined values of “yellow green to orange” and “yellow green” from Table 10. From the error rates in “Error Rate (A) [%]” in Table 9, it can be inferred that the total of “Yellow Green to Orange” and “Yellow Green” serve as the baseline for predicting harvest quantities one week later. However, the predicted harvest quantities are 20% lower than the actual harvested quantities, indicating a significant discrepancy. Therefore, certain adjustments have been made. “Analysis Results (B)” in Table 9 represents the harvest quantities obtained by adding the “Light Green to Yellow Green” values from Table 10 to “Analysis Results (A)”. The error rates after adding approximately 3–8% of the “Light Green to Yellow Green” quantities to all rows are shown in “Error Rate (B)”. By incorporating about 4% of the “Light Green to Yellow Green” values, the error rate is reduced to around 1–4%. This suggests that adding approximately 4% of the “Light Green to Yellow Green” quantities yield a more accurate estimation of harvest quantities.

Table 9. Comparison of analysis results between actual harvest and one week before harvest.

	1 Row	2 Row	3 Row	4 Row
Actual Harvest	166	188	158	109
Analysis Results (A)	152	157	124	85
Error Rate (A) [%]	8.43	16.49	21.52	22.02
Analysis Results (B)	165	186	160	110
Error Rate (B) [%]	0.60	1.06	1.27	0.92

Table 10. Analysis of harvest date obtained a week ago.

Color Type	1 Row	2 Row	3 Row	4 Row
Red	45	25	42	42
Orange to Red	13	22	31	15
Orange	20	28	20	25
Yellow Green to Orange	43	52	29	23
Yellow Green	109	105	95	62
Light Green to Yellow Green	457	577	453	622
Light Green	808	1108	1330	1085
Green to Light Green	48	53	68	46
Green	5	5	0	1

The progression of ripeness is influenced not only by the ripeness stage, which is the benchmark at harvest, but also by temperature changes and sunlight (light intensity) one week later. Predicting crop yield more accurately is possible by considering how these factors affect growth and ripeness. Therefore, it is important to utilize data that includes these conditions in future research. In this study, the analysis results were based on quantitative data, so we believe that this technology can be used in a wide range of applications in other regions and environments. However, the experimental results were based on operations at a single farm, and cultivation methods, optimal harvest times, shipping

methods, etc., may differ from country to country and region to region. Therefore, we believe that the usefulness of this engineering technology can be demonstrated by applying the harvest prediction method to the conditions of each region.

However, there were some areas where our technology was insufficient for verification, and we were unable to successfully apply the harvest prediction method to the measurement of size and volume in the harvesting procedures and shipping systems of the farmers who conducted the demonstration experiment. In the future, we will aim to achieve a more accurate harvest prediction method by further researching the relationship between ripeness and size and volume to address this issue.

The experiment confirmed that hue discrimination could be performed with high accuracy and that the hue one week before harvest could be based on “Yellow Green to Orange” and “Yellow Green”. However, further improvements in accuracy are required to minimize these errors. Additionally, although this experiment was conducted during the winter season, harvest timing is expected to be adjusted based on earlier ripening stages, such as “Light Green,” since ripening occurs earlier after harvest in the summer. The selection of appropriate harvest criteria for different seasons will be the subject of future research.

8. Summary

In this study, we propose a yield estimation technique that enables growers to determine the yield of tomatoes in advance. Additionally, we develop an AI identification camera. The camera detects tomatoes using the YOLO machine learning model and calculates their size and volume using point cloud data. It is also capable of accurately identifying the ripeness of tomatoes using hue analysis, which is independent of light intensity. Furthermore, the AI identification camera was operated in an actual field, and the results were compared with the yield data. Consequently, the error rate was confirmed as 6.85%, which is a level that can withstand practical applications.

These results allow the grower to estimate the yield in advance of harvesting and shipping and to adjust the labor and machinery operation schedule for harvesting operations in advance. In addition, it is easier to make appropriate supply plans with markets and retailers, and when a large harvest is expected and a surplus is anticipated, it is possible to reduce the risk of oversupply or shortage in relation to demand by reducing waste and deterioration losses for tomatoes when processing to avoid a decline in market prices and by arranging storage to realize a stable supply. This can reduce the risk of oversupply or shortage in relation to demand and ensure a stable supply.

The greenhouse experiment was conducted in winter, but it is anticipated that harvest timing will need to be adjusted based on earlier ripening stages, such as “Light Green,” since ripening occurs more quickly after harvest in the summer. The selection of appropriate harvesting criteria for different seasons will be the subject of future research. If these issues are resolved, growers can optimize harvest timing, improving efficiency, reducing costs, and maximizing profits.

The harvest forecasting technology developed in this study provides benefits not only to growers but also to shippers and consumers. Specifically, it is expected to ensure a stable supply of tomatoes and improve their quality, thereby contributing to the increased efficiency and sustainability of the entire agricultural industry.

Author Contributions: Experiments and overall compilation, Y.O. and T.H.; Image analysis support, K.E. and T.K.; Development of AI camera for experiments, D.H. All authors have read and agreed to the published version of the manuscript.

Funding: Not applicable. This research received no external funding

Data Availability Statement: Data are available from the authors upon request.

Acknowledgments: We would like to thank Denno of the JA Aichi Kaifu Agricultural Products Department's Bearer Division and Matsuoka, the tomato farmer, for their cooperation in carrying out the results of this research.

Conflicts of Interest: Author Daisuke Hayashi is employed by the company Kodan Electronics Co., Ltd. The remaining authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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